

Docker Lab 1

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IDA

Commands Summary:

What is the difference between:

- CMD & ENTRYPOINT

Problem 1:

- Run the container nginx `docker run -d --name my-nginx nginx`
- Check the container status `docker ps`
- Start the stopped container `docker start my-nginx`
- Remove the container `docker rm my-nginx`
- Remove the image `docker rmi nginx`

Problem 2:

- Run container ubuntu in an interactive mode `docker run -it --name my-ubuntu ubuntu`
- Run the following command in the container "echo docker" `echo docker`
- Open a bash shell in the container and touch a file named hello-docker `touch hello-docker`
the file hello-docker is deleted when the container is removed
- Stop the container and remove it. Write your comment about the file hello-docker `exit / docker stop my-ubuntu`
- Remove all stopped containers `docker container prune`

Problem 3:

- Deploy a MySQL database called app-database. Use the mysql latest image, and use the -e flag to set MYSQL_ROOT_PASSWORD to P4sSw0rd0!. The container should run in the background.
`docker run -d --name app-database -e MYSQL_ROOT_PASSWORD=P4sSw0rd0! mysql:latest`

Problem 4:

- Run the image Nginx ▪ Add html static files to the container and make sure they are accessible
- Commit the container with image name IMAGE_NAME `docker run -d --name my-webserver -p 8080:80 nginx`
`docker cp index.html my-webserver:/usr/share/nginx/html/index.html`

Problem 5:

- Create 2 nginx containers with 2 different network of type bridge, enter to one of them and use curl command to view the content of the other container.

```
docker network create my-network-1
docker network create my-network-2
docker run -d --name my-nginx-1 --network my-network-1 nginx
docker run -d --name my-nginx-2 --network my-network-2 nginx
docker network connect my-network-1 my-nginx-2
docker exec -it my-nginx-1 bash
curl http://my-nginx-2
```

Screenshots:

Problem 1:

The screenshot shows the Docker Desktop interface. At the top, it displays 'Container CPU usage' at 0.00% / 2000% and 'Container memory usage' at 16.04MB / 7.43GB. Below this is a search bar and a toggle for 'Only show running containers'. A table lists the containers, with one running container named 'my-nginx' using the 'nginx' image. The terminal window shows the following commands and output:

```
PS C:\Users\ywagi> docker run -d --name my-nginx nginx
bbb271715e0cfd5b6eb3b6256ac6399a2051e97f6d30bc4b95df2bb65e9382af
PS C:\Users\ywagi> docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS        NAMES
bbb271715e0c   nginx    "/docker-entrypoint..." 28 seconds ago Up 28 seconds 80/tcp        my-nginx
PS C:\Users\ywagi> docker start my-nginx
my-nginx
PS C:\Users\ywagi> docker rm my-nginx
Error response from daemon: cannot remove container "my-nginx": container is running: stop the container before removing or force remove
PS C:\Users\ywagi> docker stop my-nginx
my-nginx
PS C:\Users\ywagi> docker start my-nginx
my-nginx
PS C:\Users\ywagi> docker stop my-nginx
my-nginx
PS C:\Users\ywagi> docker rm my-nginx
my-nginx
PS C:\Users\ywagi> docker rmi nginx
Untagged: nginx:latest
Deleted: sha256:5aca99593157f4ae539a5dec1092a0ad8762f8e2eb1789085a13a0f5622369f6
PS C:\Users\ywagi> docker run -d --name my-nginx nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
Digest: sha256:5aca99593157f4ae539a5dec1092a0ad8762f8e2eb1789085a13a0f5622369f6
Status: Downloaded newer image for nginx:latest
70c2d6c366ce4544126889478112b4d08d1c3134b27494c0660c957244b5dc5a
PS C:\Users\ywagi>
```

Problem 2:

The screenshot shows the Docker Desktop interface. At the top, it displays 'Container CPU usage' at 0.00% / 2000% and 'Container memory usage' at 1.94MB / 7.43GB. Below this is a search bar and a toggle for 'Only show running containers'. A table lists the containers, with one running container named 'my-ubuntu' using the 'ubuntu' image. The terminal window shows the following commands and output:

```
PS C:\Users\ywagi> docker run -it --name my-ubuntu ubuntu
root@c9204edab5b6:/# echo docker
docker
root@c9204edab5b6:/# touch hello-docker
root@c9204edab5b6:/# exit
exit
PS C:\Users\ywagi> docker stop my-ubuntu
my-ubuntu
PS C:\Users\ywagi> docker container prune
WARNING! This will remove all stopped containers.
Are you sure you want to continue? [y/N] y
Deleted Containers:
c9204edab5b67fff90f0c3552b1077936f293837d6600f904fd59eb142998b4e

Total reclaimed space: 12.29KB
PS C:\Users\ywagi> docker run -it --name my-ubuntu ubuntu
root@c9204edab5b6:/#
```

Problem 3:

The screenshot shows the Docker Desktop interface. At the top, it displays 'Container CPU usage' at 70.63% / 2000% (20 CPUs available) and 'Container memory usage' at 445.4MB / 7.43GB. Below this is a search bar and a toggle for 'Only show running containers'. A table lists the running containers:

	Name	Container ID	Image	Port(s)	CPU (%)	Memory usage	Memory (%)	Disk read/write	Network I/O	Actions
☐	app-database	a16e83462f6e	mysql:latest		10.83%	445.4MB / 7.61G	5.72%	101MB / 284MB	1.17KB / 126B	

Below the table is a terminal window with the following commands and output:

```
PS C:\Users\ywagi> docker run -d --name app-database -e MYSQL_ROOT_PASSWORD=P4sSw0rd0! mysql:latest
a16e83462f6e97c9af956a49a97c70b0a28181c2a28ef411a4098aad8e6d6500
PS C:\Users\ywagi>
```

Problem 4:

The screenshot shows the Docker Desktop interface. At the top, it displays 'Container CPU usage' at 0.00% / 2000% (20 CPUs available) and 'Container memory usage' at 16.29MB / 7.43GB. Below this is a search bar and a toggle for 'Only show running containers'. A table lists the running containers:

	Name	Container ID	Image	Port(s)	CPU (%)	Memory usage	Memory (%)	Disk read/write	Network I/O	Actions
☐	my-webserver	5917052bfc97	nginx	8080:80	0%	16.29MB / 7.61G	0.21%	0B / 12.3KB	2.87KB / 1.39KB	

Below the table is a terminal window with the following commands and output:

```
PS C:\Users\ywagi> docker run -d --name my-webserver -p 8080:80 nginx
5917052bfc97b49719f44c6225ecc65888dbab3168bc19ef5dfea5e5ba76c94c
PS C:\Users\ywagi> docker cp index.html my-webserver:/usr/share/nginx/html/index.html
Successfully copied 2388 (transferred 2.05kB) to my-webserver:/usr/share/nginx/html/index.html
PS C:\Users\ywagi> docker commit my-webserver my_first_image_commit
sha256:13cd5664ea2af1589d3aabd4a48f859d39fc5c8f318e73204b0bf7fc22bf5514
PS C:\Users\ywagi> cat .\index.html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My HTML</title>
</head>
<body>
  <h1>Hello My Docker</h1>
</body>
</html>
PS C:\Users\ywagi>
```



Problem 5:

Containers [Give feedback](#)

Container CPU usage

0.00% / 2000% (20 CPUs available)

Container memory usage

33.43MB / 7.43GB

[Show charts](#)

Only show running containers

☐		Name	Container ID	Image	Port(s)	CPU (%)	Memory usag...	Memory (%)	Disk read/write	Network I/O	Actions
☐		my-nginx-1	894ca31530a0	nginx		0%	17.14MB / 7.61G	0.22%	0B / 12.3KB	3.38KB / 712B	
☐		my-nginx-2	ff56b7aabdce	nginx		0%	16.29MB / 7.61G	0.21%	0B / 12.3KB	3.16KB / 1.77KB	

Terminal

PS C:\Users\ywagi> docker network create my-network-1
21ed10422ff1cb173f5682b8b2a14723795f3246af11a947762ac0b7a4c8bcab
PS C:\Users\ywagi> docker network create my-network-2
b2a6a58ad48bd4da87e9552319cf0a4e835a353a6f8f46bf0fd832bcb4d0da
PS C:\Users\ywagi> docker run -d --name my-nginx-1 --network my-network-1 nginx
894ca31530a0e5ccaa6e5dd05daebbe87e5c3f49661f9b57c626af4877b6789
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
nginx.org.

To engage with the community please visit
community.nginx.org.

For enterprise grade support, professional services, additional
security features and capabilities please refer to
f5.com/nginx.</p>